

INVENTORY MANAGEMENT SYSTEM

C# Console Application Project

Project Type	Console Application
Language	C#
Platform	.NET / Visual Studio Code
Core Concepts	Variables, Methods, If-Else, Switch, Loops, Lists, Classes

LIVE PROJECT LINK

<https://dotnetfiddle.net/SBIRx1>

The complete project document contains requirements, objectives, design, flow, C# implementation, running instructions, and the project link.

1. Requirements and Objectives

Functional Requirements

No.	Requirement
1	Add new products with product name, price, and stock quantity.
2	Update the stock quantity of an existing product.
3	View all products with their prices and current stock levels.
4	Remove products from the inventory.
5	Provide a menu-driven console interface.
6	Validate basic input such as negative price or stock values.

Non-Functional Requirements

- The application should be simple and easy to use through the console.
- The program should give clear success, error, and validation messages.
- The code should be organized into classes and methods for readability.
- The application should run as a standard C# console project using .NET.

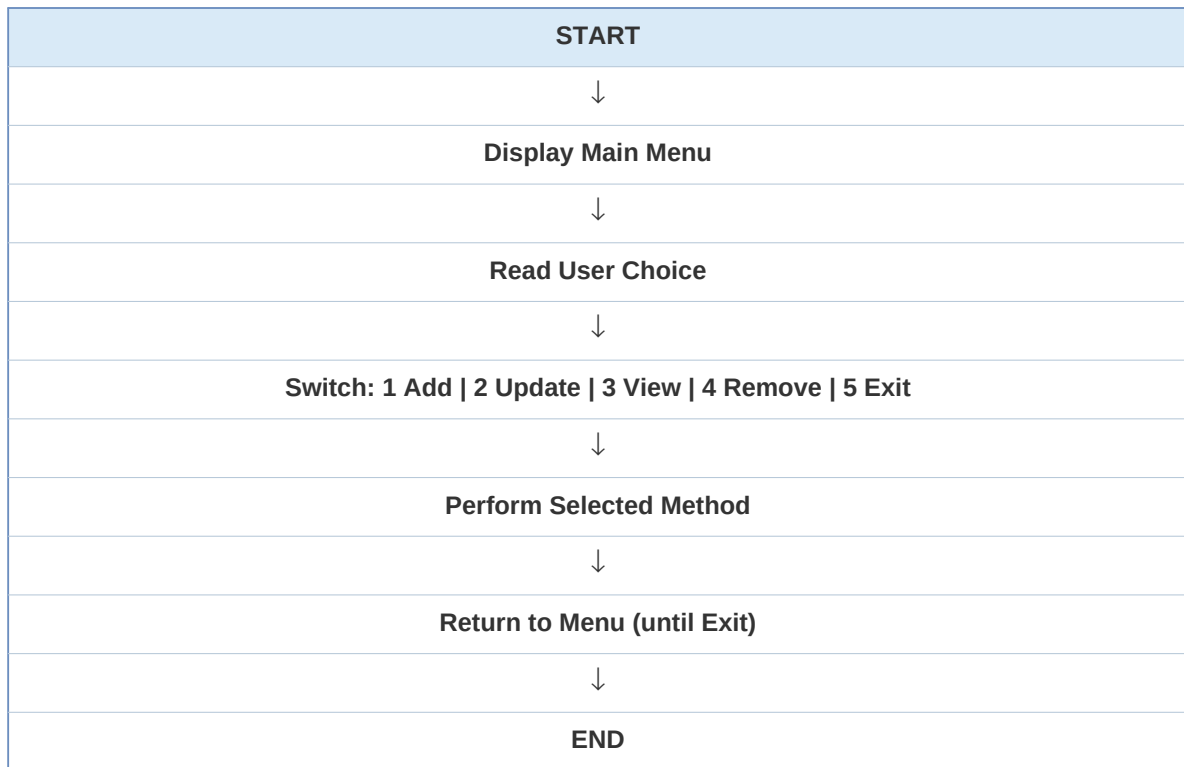
Objectives

- Maintain product information in an inventory list.
- Allow the user to add, update, display, and remove products.
- Demonstrate C# control structures such as if-else and switch.
- Demonstrate loops such as while and foreach.
- Define and call methods to divide the application into manageable tasks.

2. Design Outline

Component	Design
Product data	Name, Price, Stock
Data structure	List<Product>
Class	Product stores product information
Manager class	InventoryManager handles inventory operations
Methods	AddProduct(), UpdateStock(), ViewProducts(), RemoveProduct(), Main()
Decision structures	if-else for validation/search; switch for menu selection
Loops	while for continuous menu; foreach for displaying products

Program Flow



Key Control Structures

Requirement	Used In Code
If-else	Validates price/stock and checks whether a product exists.
Switch	Processes the user's main menu choice.
While loop	Keeps the menu running until the user chooses Exit.
Foreach loop	Displays every product in the inventory.
Methods	Separates Add, Update, View, Remove, and Main operations.

3. C# Implementation

Create a .NET console project and place the following code in **Program.cs**.

```
using System;
using System.Collections.Generic;

class Product
{
    public string Name { get; set; }
    public double Price { get; set; }
    public int Stock { get; set; }

    public Product(string name, double price, int stock)
    {
        Name = name;
        Price = price;
        Stock = stock;
    }
}

class InventoryManager
{
    static List<Product> products = new List<Product>();

    static void AddProduct()
    {
        Console.Write("Enter product name: ");
        string name = Console.ReadLine();

        Console.Write("Enter product price: ");
        double price = Convert.ToDouble(Console.ReadLine());

        Console.Write("Enter stock quantity: ");
        int stock = Convert.ToInt32(Console.ReadLine());

        if (price < 0 || stock < 0)
        {
            Console.WriteLine("Price and stock cannot be negative.");
            return;
        }

        products.Add(new Product(name, price, stock));
        Console.WriteLine("Product added successfully.");
    }

    static void UpdateStock()
    {
        Console.Write("Enter product name: ");
        string name = Console.ReadLine();

        Product product = products.Find(p =>
            p.Name.Equals(name, StringComparison.OrdinalIgnoreCase));

        if (product == null)
        {
            Console.WriteLine("Product not found.");
            return;
        }
    }
}
```

```

        Console.WriteLine("Enter new stock quantity: ");
        int newStock = Convert.ToInt32(Console.ReadLine());

        if (newStock < 0)
        {
            Console.WriteLine("Stock cannot be negative.");
        }
        else
        {
            product.Stock = newStock;
            Console.WriteLine("Stock updated successfully.");
        }
    }

    static void ViewProducts()
    {
        if (products.Count == 0)
        {
            Console.WriteLine("Inventory is empty.");
            return;
        }

        Console.WriteLine("\n--- Inventory ---");

        foreach (Product product in products)
        {
            Console.WriteLine(
                $"Name: {product.Name}, Price: {product.Price:C}, Stock: {product.Stock}");
        }
    }

    static void RemoveProduct()
    {
        Console.WriteLine("Enter product name to remove: ");
        string name = Console.ReadLine();

        Product product = products.Find(p =>
            p.Name.Equals(name, StringComparison.OrdinalIgnoreCase));

        if (product == null)
        {
            Console.WriteLine("Product not found.");
        }
        else
        {
            products.Remove(product);
            Console.WriteLine("Product removed successfully.");
        }
    }

    static void Main()
    {
        bool running = true;
    }

```

```

while (running)
{
    Console.WriteLine("\n===== INVENTORY MANAGEMENT SYSTEM =====");
    Console.WriteLine("1. Add Product");
    Console.WriteLine("2. Update Stock");
    Console.WriteLine("3. View Products");
    Console.WriteLine("4. Remove Product");
    Console.WriteLine("5. Exit");
    Console.Write("Enter your choice: ");

    string choice = Console.ReadLine();

    switch (choice)
    {
        case "1":
            AddProduct();
            break;

        case "2":
            UpdateStock();
            break;

        case "3":
            ViewProducts();
            break;

        case "4":
            RemoveProduct();
            break;

        case "5":
            running = false;
            Console.WriteLine(
                "Thank you for using the Inventory Management System.");
            break;

        default:
            Console.WriteLine("Invalid choice. Please try again.");
            break;
    }
}
}
}

```

4. Project Link

The project is available online at the following DotNetFiddle link:

LIVE PROJECT LINK

<https://dotnetfiddle.net/SBIRx1>

Click the link above to open the online C# project and run it in DotNetFiddle.

5. How to Run the Project Locally

1. Install the .NET SDK.
2. Install Visual Studio Code with C# / C# Dev Kit.
3. Create a folder named InventoryManagementSystem.
4. Create or open a .NET console project.
5. Replace Program.cs with the code in this document.
6. Run the project using **dotnet run**.

6. Project Rubric Checklist

Criteria	Evidence in Project	Points
Requirements & Objectives	Functional/non-functional requirements and objectives are documented.	5
Design Outline	Classes, variables, methods, control structures, loops, and flow are documented.	5
Control Structure	if-else and switch are implemented.	5
Loop	while and foreach loops are implemented.	5
Method	Methods are defined and called for the inventory operations.	5
Total	Complete project	25

Conclusion

The Inventory Management System is a complete C# console application demonstrating product management, methods, decision-making, and loops. The online project link is included for direct access to the runnable project.